ORACLE 2019 OLD PAPER

Q1. Attempt the following:

(1) Full form of SQL.

ANS:- SQL- Structured query language.

(2) Full form of DBMS.

ANS:- DBMS-Database management system.

(3) ER stands for ----------

ANS:- Entity relationship.

(4) Define term database.

ANS:- Collection of meaningful data is called database.

(5) Define term primary key.

ANS:- A primary key is used to ensure data in the specific column is unique.

(6) -------- is symbol for concatenation operator in sql.

ANS:- ||

(7) --------operator is used for search specific pattern from the table.

ANS:- Like

(8) Define term synonyms.

ANS:-Synonyms is an alias or an alternate name for a table, view, sequence, etc.

(9) Define Floor().

ANS:-The floor() function returns rounded nearest small value of given number.

(10) Define ceil().

ANS:- The ceil() function returns rounded nearest big value of given number.

(11) ---- command is used to display the structure of a

Table.

ANS:- describe (desc)

(12) Full form of DCL.

ANS:- DCL:-Data control language.

(13) Full form of DML.

ANS:- DML-Data manipulation language.

(14) Full form of PL/SQL.

ANS:- PL/SQL-Procedural language/Structured query language

(15) Define term Index.

ANS:- An index is a object that contains an entry for each value that appears in the indexed columns of the table and provides direct, fast access to rows.

(16) Define term commit.

ANS:- Commit is a transaction control language in sql. That permanently save all the changes made in the transaction of a database or table.

(17) Inner join also called----

ANS:- Equi join

(18) ---- command is used to display record from the table.

ANS:-Select

(19)--- command is used to edit record .

ANS:- Update

(20) DDL stands for ------

ANS:- Data definition language.

Q2 (A) Attempt any three.

(1) Explain select command.

ANS:-The SELECT command is a SQL command that is used to retrieves command.

The syntax for a SELECT command is as follows :-

SELECT column_list FROM table_name

[WHERE condition]

[ORDER BY column-name].

(2) Explain insert command.

ANS:- The INSERT command is data manipulation commands, which is used to manipulate data by inserting the information into the tables.

This command is used to add records to a table.

The syntax for a INSERT command is as follow:-

INSERT into table_name(column1,column2)

VALUES (value1, value2);

(3) Explain ALTER command.

ANS:- The ALTER command is used to add, delete or modify columns in an existing table. The ALTER command is also used to add and drop various constraints on an existing table.

The syntax for a ALTER command is as follows :-

ALTER TABLE table_name

ADD column_name datatype;

(4) Explain DELETE command.

ANS:- The DELETE command is a data manipulation command which is used to remove from a table.

The syntax for a DELETE command is as follow:-

DELETE FROM table_name

WHERE condition;

(5) Explain UPDATE command.

ANS:- The UPDATE command is a data manipulation command which is used to edit the records of a table.

It is used along with the SET clause, operationally, a WHERE clause may be used to match condition.

The syntax for a UPDATE command is as follow:-

UPDATE table_name

SET column1 = new_value1

WHERE condition;

(6) Explain Group By and Having.

ANS:- The GROUP BY statement groups rows that have the same values into summary rows.

The GROUP BY statement is often used with aggregate functions (COUNT,

MAX, MIN, SUM, AVG) to group the result-set by one or more columns.

The syntax for Group By :-

SELECT column_name

FROM table_name

GROUP BY column_name

ORDER BY column_name;

HAVING CLAUSE :-

Having clause special use for give condition to group by like WHERE.

You use the HAVING clause to filter groups of row GROUP BY can be used without HAVING.But HAVING must be used with GROUP BY.

You place the HAVING clause after your GROUP BY clause.

HAVING Syntax

SELECT column_name

FROM table_name

GROUP BY column_name

HAVING condition

ORDER BY column_name;

Q3. Attempt any three.

(1) Explain any two constrain in sql.

ANS:- constrain are set of rules used enforce data integrity.

- They don't take space in database like data.
- Even if single constrain ,entire record be rejected by oracle.

[1]Primary key:

- It can allow for single column.
- It dose not allow duplicate values.
- Null values are not allowed.
- It can identify the unique record in the table.

Example:

Create table stud

(sno number(3) primary key,

Sname varchar2(20),

City varchar2(10));

[2] foreign key:

- Foreign key is a field in a table which uniquely identifies each row of a another table.

- It is also known as reference key.
- The foreign field must be primary key .
- Child table may be duplicate and null values.

  Example:

  Create table stud

  (s1 number(3),

  S2 number(3),

  Total number(6),

  Sno number(6) references stud(sno));

(2) Explain data types in sql.

ANS:- A data type is a classification of a particular type of information or data.

- Each value manipulated by oracle hase a data type.

[1]char:

Size is 1 to maximum 2000 characters and if data is short than its specified than space padded rest of size.it means char is used for fixed –length string.

[2]varchar/varchar2:

- Both varchar or varchar2 are same . it store alphanumeric data and also store data.
- In varchar data is short than is length it can not remove extra length where as varchar2 remove extra length.
- Varchar2 used for variable-length strings.

[3]Long:

- Long datatype is large object (LOB) datatypes.
- It stored up to 2gb length.

- But restriction is you cannot use WHERE clause, group by clause, order by clause, distinct operator in select statement.

[4]Number:

- This datatype used to store negative and positive integers: fixed point or floating numbers with precision of up to 38 digits.
- A data type is a classification of a particular type of information or data.
- Each values manipulated by oracle has a data type.

(3) SQL v/s PL/SQ

SQL stands for structured query language.	PL/SQL stands for procedural language/structured query language.
Dose not have procedural capacity.	It has procedural capacity.
It is used to create application.	It is used to maintain data.
SQL is a subset of PL/SQL.	PL/SQL is a superset of SQL.
It is not a block structured language.	It is a block structured language.
SQL is a command interpreter.	PL/SQL is not a command interpreter.

(4) What is package?

ANS:- A package ia set of pl/sql statement .

- It is an object which can hold other objects like procedure, function, cursor, exception and so on.
- It is a container object and allow for all the object to bind together.

➢ Components:

[1]package specification:

- It is known as package header.
- It can contain the name of package.
- It doesn't contain any code for procedure and function.
- Object must be declare in this section before it is used.
- Syntax:
    - Create or replace package package_name
    - [IS/AS]
    - PL/SQL statement;
    - End package name;

[2] package body:

- It can contain the definition of public object that are declare in the specification.
- It can also have other object, which are private to the package.
    - Syntax:
    - Create or replace package body
    - Package_name [is/as]
    - PL/SQL statement;
    - End package;

➢ To call the package:
- The package object can be executed by two keywords like exec or call.
- Eg:
    - Exec jpack.sum;

➢ Advantages:
- 1] Package is more efficient because it is modual programming.
- 2] The scope of package is local as well as global.
- 3] Easy to understand.

(5) Explain union clause.

ANS:- The union operator is used to combine the results of two or more select queries into a single result set.

- It can Combines two or more queries and removes the duplication.
- The union operator creates a new table by placing all rows from two source tables into a single result table.

➢ Rules:
- 1] The number and the order of the columns must be the same in all queries.
- 2] The columns must also have similar data types.
- Syntax:
    - Select column_list from table_name
    - Union
    - Select column_list from table_name;
- Example:
    - Take two tables:

- Employee

=============

Emp_id	name	dept_id
1	Abc	1
2	Urm	2
3	Jfjg	2
4	Jjy	1
5	Iek	3
6	Ktk	5

- Department

===============

Dept_id	dept_name
1	it
2	hr
3	medical
4	quality

- select dept_id from employee
  union
  select dept_id from department;
- Output:
  - 1
  - 2
  - 3
  - 4
  - 5

(6) Explain sub-query.

ANS:- A subquery is a SQL query nested inside a larger query.

☐ Syntax :

```
SELECT   select_list
FROM     table
WHERE    expr operator
              (SELECT   select_list
               FROM     table);
```

☐ The subquery (inner query) executes once before the main query (outer query) executes.

☐ The main query (outer query) use the subquery result.

☐ A sub-query answers the queries that have multiple parts

☐ The sub-query answers one part of the question and the parent query answers the other part.

☐ **Types of sub-query:**

- ☐ Single row sub queries
- ☐ Multiple value sub queries
- ☐ Multiple column sub queries
- ☐ Multiple row sub queries

[1] Single row sub queries:

- It returns single value from a table.

☐ Example

- Select ename, sal from emp
- Where sal = (select max(sal) from emp);

☐ Output

- ENAME SAL
- ---------- ----------
- KING 5000

[2] Multiple value sub queries:

- ☐ This returns more than one value to the parent query statement.
- ☐ IN operator is the most commonly used in multiple row sub query
- ☐ Example
    - Select ename, sal, job from emp
    - Where sal in(select max(sal) from emp group by job);
- ☐ **Out put**
  - ENAME SAL JOB
  - ---------- ---------- ---------
  - MILLER 1300 CLERK
  - ALLEN 1600 SALESMAN
  - KING 5000 PRESIDENT
  - JONES 2975 MANAGER
  - FORD 3000 ANALYST
  - SCOTT 3000 ANALYST

[3] Multiple column sub queries:

- ☐ This returns values more than one column in subquery
- ☐ Example
    - Select ename, sal , job from emp
    - Where (sal,job) in (select max(sal),job from emp group by job);
- ☐ **Output :**
  - ENAME SAL JOB
  - ---------- ---------- ---------
  - MILLER 1300 CLERK
  - ALLEN 1600 SALESMAN
  - KING 5000 PRESIDENT
  - JONES 2975 MANAGER
  - FORD 3000 ANALYST
  - SCOTT 3000 ANALYST

[4] Multiple row sub queries:

- This type of sub-query returns more than one row of result from the sub-query
- Example
    - Select ename, deptno from emp
    - Where deptno not in (select deptno from emp where ename=‘SCOTT’);
- **Output**
    - ENAME DEPTNO
    - ---------- ----------
    - ALLEN 30
    - WARD 30
    - MARTIN 30
    - BLAKE 30
    - CLARK 10
    - KING 10
    - TURNER 30
    - JAMES 30
    - MILLER 10

(C) Attempt any two

(1) Explain Dr.E.F Codd rules.

Ans:- •Rule 1-Information Rule:-

All data should be presented in table.

•Rule 2-Guaranteed Access Rule:-

All data should be accessed without any problem.

•Rule 3-Systematic Treatment of NUll values:-

A field should be empty(null) which is different from 0(zero) or empty field, this can not apply to primary key.

•Rule 4-Dynamic online catlog on the basis of relational model:- A relational database must provide access to its structure through the same tools that are usedto access the data.

•Rule 5-Comprehensive data sublanguage rule:-

The database must be support at list one clearly defined language that includes functionality fordata definition or manipulation or integrity.

•Rule 6-View updating rule:-

Each view should be support the same full range of data manipulation that has direct access to atable available.

•Rule 7-High-level insert, update, delete:-

This rule states that insert, update, delete should be supported for any retrievable rather than just for a single row in a single table.

•Rule 8-Physical data independence rule:-

The user should remain isolated from the physical method of sorting and retrieving information from the database.

•Rule 9-Logical data independence:-

How data is accessed should not be changed when the logical structure of database changes.

•Rule 10-Integrity independence:-

The database language like SQL should support all types of user condition.

•Rule 11-Disrtibution independence:-

It is possible that the data can be distributed easily.

•Rule 12-Non-subversion rule:-

There should be no way to modify the database structure other than the multiple row database language like SQL.

(2) Explain IN, ANY, BETWEEN in SQL.

ANS:-

IN:-

•IN is an operator in SQL, which is generally used with a where clause.

•Using IN operator, multiple values can be specified.

•It allows us to easily test if an expression matches any value in a list of values.

•IN operator used to replace many or conditions.

Syntax

Select column_name from table_name

Where column_name in (value1, value2....);

Any:-

•The ANY operator compares the given value to each subqurery value and returns those values

that satisfy the condition.

•It always evaluates to TRUE if at least one sub query value matches according to the given condition.

Syntax

Select column_name from table_name

Where column_name any (select column_name from table_name where[condition]);

ALL:-

•ALL operator is used with select, where and having statements. •ALL means that the condition will be true only if the operation is true for all values in the range.

Syntax

Select column_name from table_name

Where column_name all (select column_name from table_name Where[condition]);

BETWEEN:-

•The BETWEEN operator selects values within a given range. The values can be numbers, text, or dates.

•The BETWEEN operator is inclusive: begin and end values are included.

Syntax

Select column_name from table_name

Where column_name between value1 and value2;

(3) Explain Index.

Ans:- An index is an object that contains an entry for each value that appears in the indexed column of the table and provides direct, fast access to rows.

Types of index:- a)B-tree index

•B-tree is a self-balancing tree data structure that keeps data stored and allows searches, sequential access, insertion, and deletions in time.

•Each page in an index B-tree is called an index node. The top node of the B-tree is called the root node. The bottom nodes in the index are called the leaf nodes.

b)Bitmap index

•A bitmap is a mapping from one system such as integers to bits. It is also known as bitmap

index or a bit array.

•The memory is divided into units for bitmap.

•These units may range from a few bytes to several kilobytes.

•The bitmap provides a relatively easy way to keep track of memory as the size of the bitmap is only depend on the size of the memory and the size of units.

c) Functional based index

•Function-based indexes allows you to create an index based on a function or expression.

•Function-based indexes can involve multiple columns, arithmetic expression, or may be a pl/sql function.

Syntax

Create index column_name on table_name(upper(column_name));

d) Application domain indexes

•Application domain indexes are custom indexes that are specific to the needs of your application.

•They can be based on the semantics of your data and queries.

(3) Explain commit, rollback, savepoint.

Ans:- Commit:- Commit is a transaction control language in SQL. It lets a user permanently save all the changes made in the transaction of a database or table. Once you execute the commit, the database cannot go back to its previous state in any case.

Syntax

Commit;

Rollback:-

Rollback is a transactional control language in SQL. It lets a user undo those transactions that aren't saved yet in the database. One can make use of this command of they wish to undo Any changes or alterations since the execution of the last commit. Syntax

rollback;

Savepoint:-

A savepoint is a point in a transaction when you can roll the transaction back to a certain point without rolling back the entire transaction.

Syntax

Savepoint savepoint_name;

This command servers only in the creation of a savepoint among all the transactional

statements. The rollback command is used to undo a group of transactions. Rollback to savepoint_name;

(5) Explain PL/SQL block structure in detail.

ANS:- In PL/SQL, all statements are classified into units that is called blocks. PL/SQL blocks can include variables, conditional statements and exception handling. Blocks can also build a function or a procedure or a package.

The declaration section:-

Code block start with a declaration section, in which memory variables, constants, cursors and other oracle objects can be declared and if required initialize.

The begin section:-

Consist of set of SQL and PL/SQL statements, which describe processes that have to be applied to table data. Actual data manipulation, retrieval, looping and branching constructs are specified in this section.

The exception section:-

The section deals with handling errors that arise during execution data manipulation statements, which make up PL/SQL code block. Errors can arise due to syntax, logic or validation rule.

The end section:-

This marks the end of PL/SQL block.

Example:- declare

```
y number(5);
bcheck exception;
begin
y:=&y;
If y<18 then
raise bcheck;
else
dbms_output.put_line('you can vote');
end if;
exception
when bcheck than
dbms_output.put_line('you cannot vote');
end;
```

3 (A) Attempt any three

(Q1) Explain Import and Export.

ANS:- Exporting data is the process of converting data from a database into one of many interchange data formats so that the data can be used by systems that do not interface directly with the database or to be imported in to another database.

Importing data is the process of taking data presented in some interchange format and converting it for use within a database system.

(Q2) What is view with example?

ANS:- A view is a virtual table that stores only definition in the form of SQL statement. It is used to hide information from other user. Syntax

Create [or replace][force] view < view _name> as

Select column_name from table_name Where[condition];

Example

Create table student

(Sno number(3), Sname varchar2(20), Salary number(5));

Create or replace view v1 as

Select salary from student Where salary>1000

With read only;

(Q3) Explain lock in concurrency control.

ANS:- A Lock is a variable assigned to any data item in order to keep track of the status of that data item so that isolation and non-interference is ensured during concurrent transactions.

(Q4) Explain Grant command.

ANS:- SQL Grant command is specifically used to provide privileges to database objects for a user. This command also allows users to grant permissions to other users too.

Syntax:

grant privilege_name on object_name to {user_name | public | role_name}

Here privilege_name is which permission has to be granted, object_name is the name of the database object, user_name is the user to which access should be provided, the public is used to permit access to all the users.

(Q5) Explain Revoke command.

ANS:- Revoke command withdraw user privileges on database objects if any granted. It does operations opposite to the Grant command. When a privilege is revoked from a particular user U, then the privileges granted to all other users by user U will be revoked.

Syntax:

revoke privilege_name on object_name from {user_name | public | role_name}

(Q6) What is redo logs files?

ANS:- •Redo logs files are operating system files used by oracle to maintain logs of all transactions performed against the database.

•The primary purpose of these log files is to allow oracle to recover changes made to the database in the case of a failure. •During crash recovery, the redo log is crucial for replaying the logged changes to bring the database to a consistent state.

•Redo log files are created when the database is created.

3(b)Attempt any three

(1) Explain SQL *loader.

ANS: SQL loader utility is used to load data from other data source into oracle.

* SQL loader provides several methods to load data.

*SQL loader is uses in two ways with or without control file.

*Features of SQL loaders:

1.Load data from multiple data files during the same load session.

2.load data into multiple tables during the same load session.

3.Specify the character set of data.

4.Manipulate the data before loading it , using SQL functions.

5.Load data from data.

6.Selectively load data.

(2) Differentiate: function v/s procedure

ANS:

Function	Procedure
Functions take parameters and return a value.	Procedure does not take any parameters and does not return value.
Function does not allows DML statements.	Procedure allows DML statements.
A function has a return type.	A procedure does not have return type.
Functions can be called from procedure.	Procedure cannot be called from a function.
We cannot use transaction in function.	We can use transaction in procedure.

(3) Explain varray.

ANS:-Varray stand for "variable size array".

* The varray is one of the three types of collection in pl/sql.

*The Varray's key feature is that when you declare varray type,you specify the maximum number of elements that can be defined in the varray.

* A varray type is created with the create type statement.

* A varray is single-dimensional collectios of elements with the same data type.

* Varray is used more frequently when the size of the array is known to us.

* You must specify the maximum size and the type of elements.

*Syntax:-

Create or replace type type_name is varray (n) of data_type;

* 'n' is the number of elements (maximum) in varray.

(4) Explain sequence with appropriate example.

ANS: A sequence is a set of integers that are generated in order on demand.

* Sequence are frequently used in many databases.

* All sequence attributes are optional and they all have default values.

* Each sequence has two columns:NEXTVAL and CURRVAL.

* NEXTVAL is used for next value while CURRVAL is used for current value of data in sequence.

* Sequence can be created,changed and dropped with the following three SQL commands.

1 Create Sequence

2 Alter Sequence

3 Drop Sequence

* Syntax:-Create sequence sequence_name

Start with <no>

Increment by <no>

Minvalue <no>

Maxvalue <no>

Cycle

Cache <no>;

* Example:

Create Sequence s1

Start with 5

Increment by 5

Minvalue 1

Maxvalue 20

Cycle

Cache 2;

Select s1.nextval from dual;

(5) Explain %type and %rowtype

ANS: %TYPE:-

* %type is used to declare the variable of the same data type as the database column.

* This is called dynamic data type assignment.

* Changes in table data type direct reflected in the program.

* Syntax:-Variable table_name.field_name%type;

%ROWTYPE:-

* %rowtype is used to fetch the entire rows of the table.

* With the help of this,we don't have to describe each field in the record definition.

* It means oracle select data type automatically.

* Syntax:-

Variable table_name%rowtype;

(6) Explain exception handling in PL/SQL. ANS:-

* An exception is an error which disrupts the normal flow of program instructions.

* PL/SQL provides exception block which raises the exception thus helping to find out the fault and resolve it.

* There are two type of exceptions:-

1 System Defined Exception

2 User Defined Exception

* No_data_found:

* It is a user defined exception.

* Which is generally use for select statements,when no record match we can use this exception.

* To create user defined exception just follow this steps:

    1 Declare exception in declaration part.

    2 Raise declared exception in execution part.

    3 In last exception part put your declared exception in between "when" and "then" for proper message.

3(c) Attempt any three :

(1) Explain package with example.

ANS:- A package is an object, which holds other objects.Like procedure, functions, cursor, exception etc.

-> Components of package

- Package specification
- Package body

--> package specification :-

- Contains name of the package.
- Contains declaration of procedure, function, cursors, exception etc.
- It does not contain any code for procedures/functions.

--> syntax of specification of package:-

create [or replace] package <package name> {as/is}

<statements>

End [package name];

--> package body :-

- Package body contains the definition of public objectss that are declared in the specification.
- Private objects are defined in package body only and they are not visible/accessible outside the package.
- The variables declared in the package specification can be accessed by any procedure/function within the package.
- Package object referred as

- exec <package name>.<object name>;

--> package body syntax:-

create [or replace] package body <package name> {as/is}

<statements>

End [package name];

--> Example:-

-- create package:-

Create or replace package jpack as

procedure jemp(jno emp.empno%type);

end jpack;

-- create package body:-

Create or replace package body jpack as

Procedure jemp(jno emp.empno%type) as

Jname emp.ename%type;

```
Begin

Select ename into jname from emp where empno=jno;

dbms_output.put_line('Employee '||jname||' found');

End;

End jpack;

/

-- Execute:-

exec jpack.jemp(7787);
```

(2) Explain triggers with example.

ANS:- Triggers are the programs that are executed automatically in response to a Change in database.

- Oracle allows special types of procedures that are automatically executed when events like insert/update/delete occurs. These event procedures called 'Database Triggers'.

- You may not use commit, rollback and savepoint statement within trigger blocks.

- : old and :new are concept of Trigger.

- The :new refers to a record containing the new, post change data.

- The :old refers to a record containing the old, pre-change data.

--> Types of Triggers:-

    - Before update statement level
    - After insert row level
    - Before delete row level

--> Trigger predicates :-

    - Inserting :- True if the trigger statement is Insert

- Deleting :- True if the trigger statement is Delete
- Updating :- True if the trigger statement is update

--> syntax of Trigger :-

create [or replace] trigger <trigger name>

[Before/after/instead of]

[Delete/insert/update [of <col1>, <col2>...]]

On <table/view name>

[For each row [when <condition>]]

Declare

<Variable>

Begin

<pl/sql statement>

Exception

<exception>

End;

--> Example :-

create or replace trigger jmsg

before insert or delete or update

On dual

For each row

Begin

If inserting then

dbms_output.put_line('record inserted in dual');

ElsIf deleting then

dbms_output.put_line('record deleted in dual');

ElsIf updating then

dbms_output.put_line('record updated in dual');

End if;

End;

(3) Explain backup & recovery in Oracle.

ANS:-

Backup :-

- A backup is a copy of data. This copy can include important part of database such as the control file and data files.
- A backup is a safeguard against unexpected data loss and application errors.
- If you loss the original data, then you can reconstruct it by using a backup.

--> Types of backup :-

- Backup of oracle data are either physical or logical.

1. Physical oracle backups :-

- Cold backup
- Hot backup
- Control file backup
- Redo log file backup

2. Logical backup :-

 Using export/ import

--> Features of backup :-

 It makes the process of data recovery simple and easy.

 It provides data security to the users.

 It is a cost-effective process to retrieve the data.

Recovery :-

 Recovery is use to store a backup of control file and reconstruct it and make it available to the oracle database server.

--> Features of recovery :-

 The process of recovering is expensive.

 When there is failure, it refers to recovering the lost data.

 It increases the database reliability.

(4) Explain cursor.

ANS:- when you execute a SQL statement from PL/SQL, the oracle assigns a private work area for that SQL statement is known as a cursor.

 The pl/SQL is mechanism by which you can name that work area and manipulate the information within it.

 Data stored in the cursor known as Active Data Set (ADS).

--> Types of cursor :-

1. Implicit cursor :- The implicit cursor are automatically generated by oracle while an SQL statement is executed.

2. Explicit cursor :- A cursor can also be opened for processing data through pl/SQL block such as user-defined cursor known as explicit cursor.

--> Attributes of cursor :-

 %isopen - returns true if cursor open, otherwise false.

 %found - returns true if record was fetched successfully, otherwise false.

 %notfound - returns true if record was not fetched successfully, otherwise false.

 %rowcount - returns number of records processed from the cursor.

--> Steps :-

1. Declaring the cursor :-

ex :-

cursor c_customers is select id, name from customers;

2. Opening the cursor :-

ex :-

open c_customers;

3. Fetching the cursor :-

ex :-

fetch c_customers into c_id, c_name;

4. Closing the cursor :-

ex :-

close c_customers;

--> Example :-

Declare

j_nam emp.ename%type;

j_sal emp.sal%type;

Cursor j_emp is select ename, sal from emp;

begin

open j_emp;

```
fetch j_emp into j_nam, j_sal;

dbms_output.put_line('salary of '||j_nam||' is '||j_sal);

close j_emp;

end;
```

(5) Explain Instance Architecture.

ANS:- when a database is started on a database server, Oracle allocates a memory area called the System Global Area (SGA) and starts one or more oracle processes.

 This combination of SGA and Oracle processes is called an oracle Instance.

--> The Instance and The Database :-

 After starting an instance, oracle associates the instance with the specified database. This is called mounting the database.

 The database then ready to be opened, which makes it accessible to authorized users.

 Only the database administrator can start up an instance and open the database.

 If a database is open, then the database administrator can shutdown is controlled through connection to oracle with administrator previlleges.

 Normal users do not have control over the current status of an oracle database.

--> An oracle Instance is build up by considering following aspects :-

1. Database process

2. Memory structure

3. Data files

1. Database process :-

 The oracle RDBMS uses two types of processes, user process and oracle process (called background process).

1. User process

2. Oracle process

 Server process

 Background process

2. Memory structure :-

 Oracle uses memory to store information such as :

 Program code.

 Information about connected session.

 Information needed during program execution.

3. Data files :-

 The oracle architecture are the physical files where our information stored on disk.

 Oracle has several data files :-

1. Database data files

2. Control files

3. Online redo log files

4. Other database related files.